

## TULCEA CATALOI, Romania

WMO: 153350

Lat: 45.1906N

Lon: 28.8242E

Elev: 6

StdP: 101.26

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -11.6      | -9.0 | -14.9                         | 1.0 | -10.3 | -11.9 | 1.3 | -7.7 | 12.2                  | -0.4 | 10.2 | 1.4  | 3.3                   | 0    | 0.470 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.3                   | 33.1            | 22.0 | 31.6 | 21.5 | 30.1 | 21.0 | 24.0                | 29.6 | 23.1 | 28.5 | 22.3 | 27.7 | 3.0                  | 90   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 22.4                            | 17.1 | 26.4 | 21.5 | 16.1 | 25.7 | 20.5 | 15.2 | 24.9 | 72.1          | 29.8 | 68.5 | 28.4 | 65.4 | 27.6 | 30.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 9.9           | 8.2  | 7.1 | DB                         | -15.7 | 35.6               | 2.7 | 2.1                                                | -17.6 | 37.1       | -19.2 | 38.4       | -20.7 | 39.5       | -22.6 | 41.0 | (4) |
| (5)               |      |     | WB                         | -16.0 | 26.0               | 2.6 | 1.5                                                | -17.8 | 27.1       | -19.4 | 28.0       | -20.8 | 28.8       | -22.7 | 29.9 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 12.2   | 0.0   | 2.1   | 6.3   | 11.7  | 17.4  | 21.6  | 23.7  | 23.3  | 18.2  | 12.3  | 7.1   | 1.9   |  |  |
|                                                                        | DBStd       | 9.18   | 5.03  | 5.12  | 4.36  | 3.50  | 3.26  | 2.89  | 2.34  | 2.91  | 3.37  | 4.07  | 4.65  | 4.70  |  |  |
|                                                                        | HDD10.0     | 1071   | 312   | 223   | 128   | 21    | 1     | 0     | 0     | 0     | 0     | 23    | 112   | 252   |  |  |
|                                                                        | HDD18.3     | 2743   | 569   | 454   | 374   | 200   | 57    | 6     | 1     | 1     | 43    | 190   | 338   | 509   |  |  |
|                                                                        | CDD10.0     | 1867   | 0     | 2     | 13    | 73    | 229   | 348   | 423   | 412   | 246   | 95    | 23    | 2     |  |  |
|                                                                        | CDD18.3     | 497    | 0     | 0     | 0     | 2     | 27    | 104   | 166   | 156   | 39    | 4     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 4521   | 0     | 0     | 2     | 26    | 231   | 881   | 1531  | 1503  | 321   | 27    | 1     | 0     |  |  |
|                                                                        | CDH26.7     | 1572   | 0     | 0     | 0     | 5     | 41    | 273   | 573   | 606   | 72    | 2     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 2.8    | 2.9   | 3.2   | 3.6   | 3.3   | 2.9   | 2.7   | 2.4   | 2.3   | 2.5   | 2.5   | 2.8   | 2.8   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 443    | 31    | 22    | 30    | 32    | 42    | 52    | 44    | 35    | 43    | 38    | 37    | 38    |  |  |
|                                                                        | PrecMax     | 648    | 96    | 61    | 73    | 66    | 100   | 94    | 142   | 116   | 124   | 104   | 91    | 99    |  |  |
|                                                                        | PrecMin     | 260    | 1     | 2     | 1     | 9     | 7     | 4     | 0     | 6     | 2     | 4     | 3     | 2     |  |  |
|                                                                        | PrecStd     | 95     | 24    | 15    | 19    | 14    | 26    | 25    | 32    | 26    | 30    | 26    | 28    | 27    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 13.6  | 17.3  | 22.0  | 26.8  | 30.2  | 33.1  | 35.1  | 35.6  | 31.0  | 26.2  | 21.8  | 15.2  |  |  |
|                                                                        |             | MCWB   | 9.3   | 11.3  | 12.2  | 17.0  | 19.6  | 22.5  | 22.1  | 22.7  | 19.4  | 18.0  | 15.3  | 11.2  |  |  |
|                                                                        | 2%          | DB     | 10.3  | 13.8  | 18.2  | 22.7  | 27.5  | 31.2  | 32.9  | 33.6  | 28.7  | 23.3  | 18.1  | 12.5  |  |  |
|                                                                        |             | MCWB   | 7.5   | 9.3   | 10.8  | 14.7  | 19.0  | 21.3  | 22.2  | 22.1  | 19.4  | 16.5  | 14.2  | 9.8   |  |  |
|                                                                        | 5%          | DB     | 8.2   | 11.2  | 15.6  | 20.3  | 25.9  | 29.7  | 31.3  | 31.8  | 26.8  | 21.1  | 15.7  | 10.3  |  |  |
|                                                                        |             | MCWB   | 6.3   | 7.8   | 9.9   | 13.5  | 18.0  | 20.9  | 21.8  | 21.4  | 18.7  | 15.9  | 12.9  | 8.3   |  |  |
|                                                                        | 10%         | DB     | 6.5   | 8.8   | 13.0  | 18.2  | 24.2  | 28.1  | 29.9  | 30.1  | 25.0  | 18.9  | 13.7  | 8.4   |  |  |
|                                                                        |             | MCWB   | 5.2   | 6.5   | 8.4   | 12.3  | 17.2  | 20.1  | 21.2  | 20.9  | 18.2  | 14.9  | 11.6  | 6.8   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 9.8   | 11.9  | 13.3  | 17.2  | 21.8  | 24.5  | 25.0  | 25.6  | 22.4  | 20.1  | 16.1  | 12.0  |  |  |
|                                                                        |             | MCDB   | 12.4  | 16.2  | 19.7  | 25.2  | 27.0  | 29.8  | 30.9  | 32.0  | 26.5  | 23.6  | 19.7  | 14.1  |  |  |
|                                                                        | 2%          | WB     | 7.9   | 9.8   | 11.5  | 15.6  | 20.1  | 23.1  | 23.9  | 24.1  | 21.1  | 17.8  | 14.6  | 10.2  |  |  |
|                                                                        |             | MCDB   | 9.8   | 13.3  | 17.1  | 21.2  | 25.7  | 28.5  | 29.2  | 30.1  | 25.4  | 21.2  | 17.6  | 12.0  |  |  |
|                                                                        | 5%          | WB     | 6.6   | 8.2   | 10.2  | 14.3  | 19.0  | 22.0  | 23.1  | 23.0  | 20.1  | 16.7  | 13.3  | 8.6   |  |  |
|                                                                        |             | MCDB   | 8.0   | 10.6  | 14.8  | 19.3  | 24.2  | 27.4  | 28.5  | 28.3  | 24.2  | 20.2  | 15.4  | 10.3  |  |  |
|                                                                        | 10%         | WB     | 5.2   | 6.8   | 8.9   | 13.2  | 17.9  | 21.1  | 22.3  | 22.1  | 19.1  | 15.4  | 12.0  | 7.0   |  |  |
|                                                                        |             | MCDB   | 6.3   | 8.6   | 12.3  | 17.3  | 22.6  | 26.3  | 27.8  | 27.9  | 23.5  | 18.3  | 13.6  | 8.3   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 6.4   | 7.5   | 8.9   | 10.2  | 10.8  | 11.0  | 11.3  | 12.1  | 11.2  | 9.9   | 7.8   | 6.7   |  |  |
|                                                                        |             | MCDBR  | 9.3   | 11.6  | 13.3  | 13.5  | 13.3  | 13.1  | 13.7  | 14.2  | 13.7  | 13.1  | 10.6  | 9.6   |  |  |
|                                                                        |             | MCWBR  | 6.6   | 7.6   | 7.4   | 6.9   | 6.2   | 5.6   | 5.7   | 5.5   | 6.3   | 7.0   | 6.5   | 6.9   |  |  |
|                                                                        |             | MCDBR  | 8.4   | 10.8  | 12.2  | 12.3  | 12.0  | 11.7  | 12.2  | 12.7  | 12.3  | 11.8  | 9.7   | 9.1   |  |  |
|                                                                        | 5% WB       | MCWBR  | 6.4   | 7.4   | 7.2   | 6.8   | 6.3   | 5.8   | 5.8   | 5.7   | 6.5   | 6.9   | 6.3   | 6.9   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.344 | 0.373 | 0.406 | 0.440 | 0.418 | 0.418 | 0.436 | 0.445 | 0.414 | 0.388 | 0.364 | 0.343 |  |  |
|                                                                        | taud        |        | 2.297 | 2.233 | 2.193 | 2.125 | 2.270 | 2.300 | 2.260 | 2.224 | 2.281 | 2.325 | 2.332 | 2.321 |  |  |
|                                                                        | Ebn at Noon |        | 744   | 789   | 816   | 819   | 853   | 852   | 830   | 806   | 800   | 765   | 713   | 701   |  |  |
|                                                                        | Edh at Noon |        | 82    | 106   | 126   | 146   | 131   | 128   | 131   | 131   | 114   | 96    | 80    | 74    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1.22  | 2.02  | 3.29  | 4.79  | 6.06  | 6.47  | 6.45  | 5.78  | 4.19  | 2.62  | 1.40  | 1.04  |  |  |
|                                                                        | RadStd      |        | 0.20  | 0.28  | 0.31  | 0.44  | 0.39  | 0.37  | 0.34  | 0.30  | 0.34  | 0.24  | 0.20  | 0.17  |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | +0.67 | N/A     | N/A    | +0.89   | N/A   | N/A   | -106        | N/A     | +118    | +83     |
| (47) Regional (29 neighbors) |  | +0.52 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +100    | +67     |

Nomenclature: See separate page